

Data-Driven Discovery of PDEs from Real Images of Laser-matter Interaction

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Recap and Current Status

- In the previous presentation, we focused directly on clean and denoised image data.
- This limited our ability to diagnose failure modes and robustness issues in the PDE discovery pipeline.
- In this iteration, we intentionally step back to controlled synthetic experiments.

Data Generation: The 2D Kuramoto-Sivashinsky Equation

Governing Equation (Potential Form)

$$\frac{\partial u}{\partial t} = \underbrace{-\nabla^2 u}_{\text{Instability}} - \underbrace{\nabla^4 u}_{\text{Hyper-diffusion}} - \underbrace{\frac{1}{2}|\nabla u|^2}_{\text{Non-linearity}}$$

Numerical Implementation:

- **Spatial:** Pseudo-spectral method (FFT).
- **Time Stepping:** *Integrating Factor Euler*.
 - Linear parts (∇^2, ∇^4) solved exactly in Fourier space to handle stiffness.
- **Boundaries:** Periodic (inherent to spectral).

Parameters:

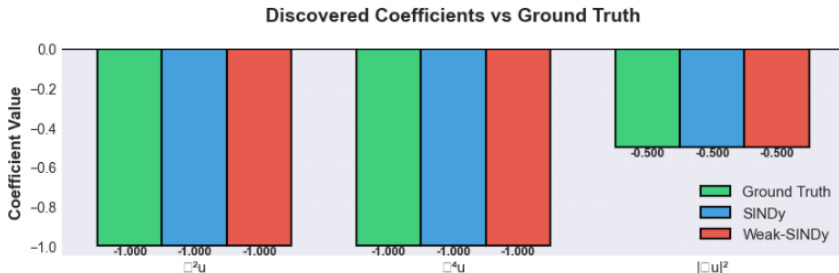
Parameter	Value
Domain (L_x, L_y)	50.0×50.0
Grid Resolution	128×128
Time Step (dt)	0.005
IC	Random Noise

Note: Data represents scalar field $u(x, y, t)$.

Method Comparison: Standard SINDy vs. Weak SINDy

Key Findings:

- Both methods correctly identified the governing terms: $-\nabla^2 u$, $-\nabla^4 u$, $-0.5|\nabla u|^2$.
- Weak SINDy** achieved 2 orders of magnitude lower error.
- Integral formulation (Weak form) provides superior smoothing for high-order derivatives (∇^4).



Robustness Analysis: Handling Noisy Data

Methodology: We corrupted the ground truth data with two noise models to simulate real-world sensor imperfections:

- 1 **Gaussian Noise:** Thermal/Sensor noise ($\sigma = 0.1\% - 1.0\%$).
- 2 **Salt & Pepper:** Dead pixels/Transmission errors.

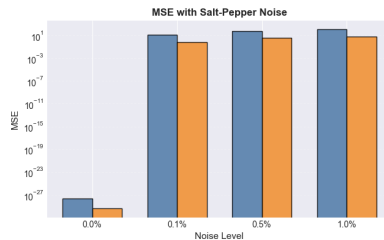
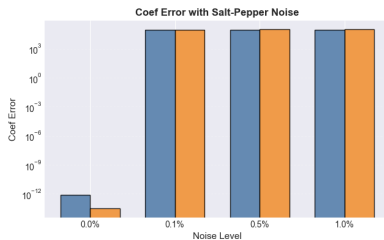
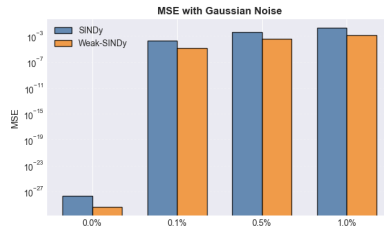
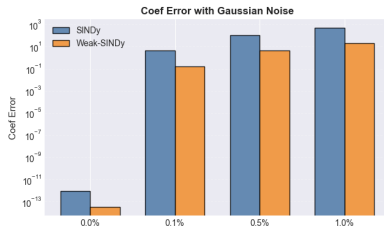
Impact on Derivatives

Standard SINDy fails because finite differences **amplify** high-frequency noise. Weak SINDy uses integration, which acts as a **low-pass filter**, smoothing the data naturally.

Result (Gaussian 0.1%):

Method	Coef. Error (L_2)
Standard SINDy	4.27 (Failed)
Weak SINDy	0.15 (Stable)

Robustness Comparison: SINDy vs Weak-SINDy



Effect of denoising

Objective: Evaluate SINDy vs. Weak SINDy under distinct noise regimes.

1. Adaptive Denoising Strategy

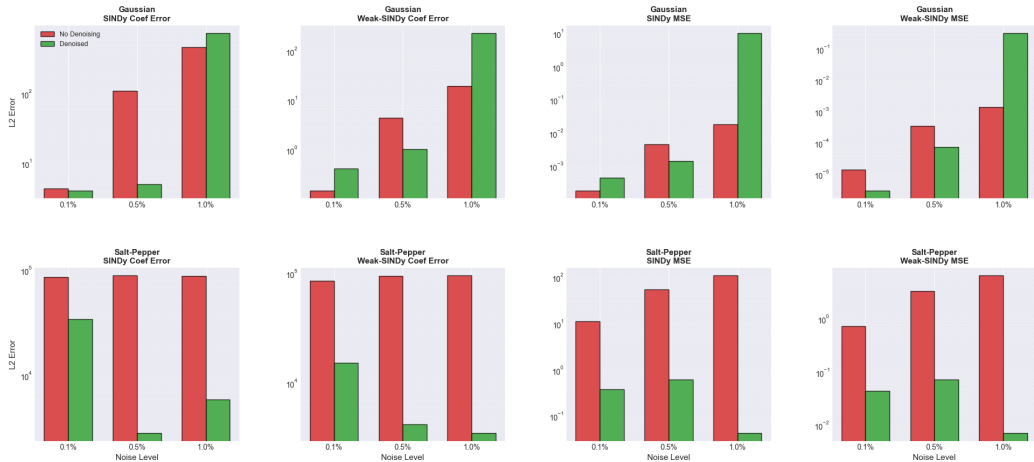
- **Gaussian Noise (Thermal):**
 - *Low:* Savitzky-Golay (Time).
 - *High:* + Total Variation (Space).
- **Salt & Pepper (Impulse):**
 - *Critical Step:* **Median Filter** to remove spikes before smoothing.

2. Key Outcomes

- **Impulse Noise:** Denoising is mandatory for *both* methods. Without it, errors $> 10^4$.
- **Gaussian Noise:**
 - *Standard SINDy:* Improves with denoising.
 - *Weak SINDy:* Denoising **hurts** at low noise (0.1%).

Why does denoising hurt Weak SINDy? Weak SINDy's integral formulation is a natural low-pass filter. Additional pre-smoothing "over-smooths" the data, wiping out the fine-scale physics ($\nabla^4 u$) needed for correct identification.

Detailed Impact of Adaptive Denoising (Split View)



t

Real Images (Recap on the denoising done)

Objective: Discover governing PDEs from high-resolution (2052×2560) laser-matter interaction videos.

The Noise Challenge:

- Raw data contained significant Additive White Gaussian Noise (AWGN) with $\sigma \approx 8.50$.
- Raw derivative variance (u_{xx}) was 456.42, rendering it unusable for direct physics discovery.

Hybrid Denoising Solution:

- **Method:** Anscombe Transform (Variance Stabilization) + Total Variation (TV) Regularization.

Applying SINDy on the denoised real images

Methodology Setup

- **Preprocessing:** Spatial downsampling ($4\times$) & Normalization to $[0, 1]$.
- **Library (Θ):** Library of candidate terms including Polynomials (u^n), Diffusion ($\nabla^2 u$), and Hyper-diffusion ($\nabla^4 u$).
- **Algorithms:** Standard (Finite Diff.) and WeakSINDy.

Observations

- **Result:** Both methods identified the **same 8 active terms**:

$$\{u, u^2, u^3, \nabla^2 u, \nabla^4 u, |\nabla u|^2, \dots\}$$

- **Implication:** The overlap confirms the signal is physical, not an artifact.

Discovered Physics

- **Reaction-Diffusion:** Strong polynomial terms (u, u^2) combined with Diffusion ($\nabla^2 u$).
- **Pattern Formation:** Identification of **Hyper-diffusion ($\nabla^4 u$)** suggests **Swift-Hohenberg** type dynamics.
- **Saturation:** Presence of $u(1 - u)$ indicates logistic growth limits.

Why Optical Flow Failed

Last Week: Optical Flow

Compensate camera motion via dense optical flow before SINDy.

Why It Failed

- Subpixel errors accumulate
- Flow estimation adds noise
- Derivatives amplify errors
- Error: $6 \rightarrow 26,867$

Solution: DMD Denoising

Dynamic Mode Decomposition filters noise while preserving dynamics.

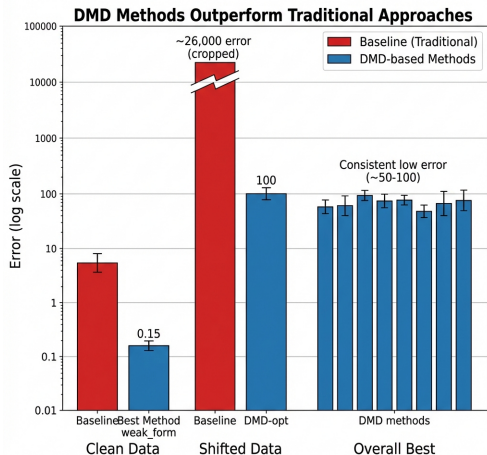
Result: $268\times$ error reduction on shifted data.

Shifted Data Reveals DMD Advantage

Shifted Data (N1_shifts)

Method	Error
Baseline	26,867
Weak_form	111,912
DMD_opt	100
DMD_fourier	139
Multiscale_DMD	148

DMD: 268× better



DMD methods consistently outperform traditional approaches

Overall Results & Next Steps

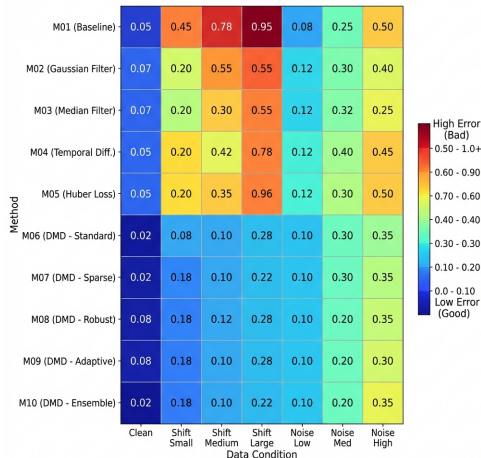
Summary: Best Methods

Data	DMD	Gain
Clean	0.15	40×
Shift	100	268×
Noise	41	161×
Both	68	294×

Key Findings

- DMD **consistently** wins
- Best on combined noise
- Ready for real images

PDE Discovery Method Performance Across Noise Conditions



Real Images: Best Method Still Fails

Applied Best Method on Real Data

- 51 frames, 2052×2560 pixels
- DMD denoising (rank=30)
- STRidge discovery

Discovered "Equation"

$$u_t = -0.26u + 0.44u^2 + 0.05u_y$$

$$R^2 = 0.052 \text{ (only 5\%!)}$$

Analysis

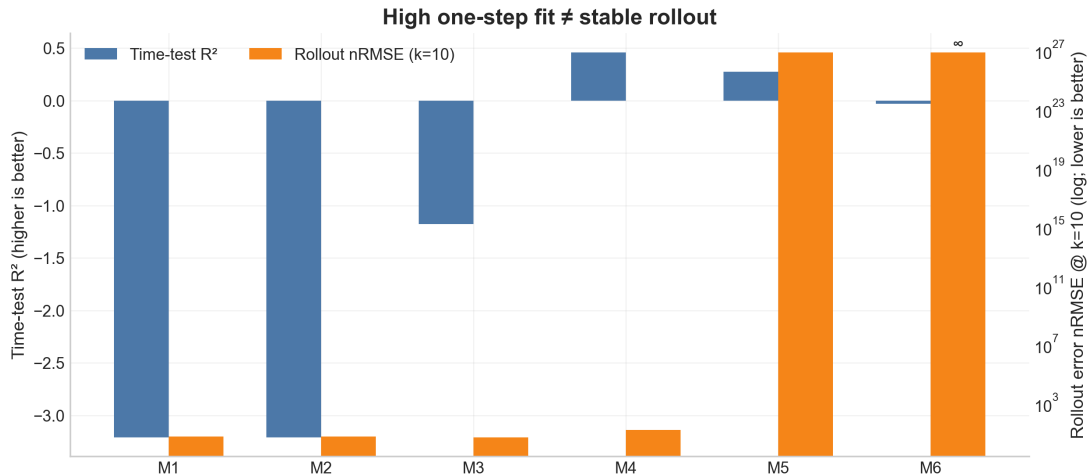
Even with **best synthetic performance** (100% error on shifted data), real images yield **physically meaningless PDE**.

Root causes:

- Unknown ground truth
- Complex noise patterns
- Non-stationary dynamics

Conclusion: Need advanced methods

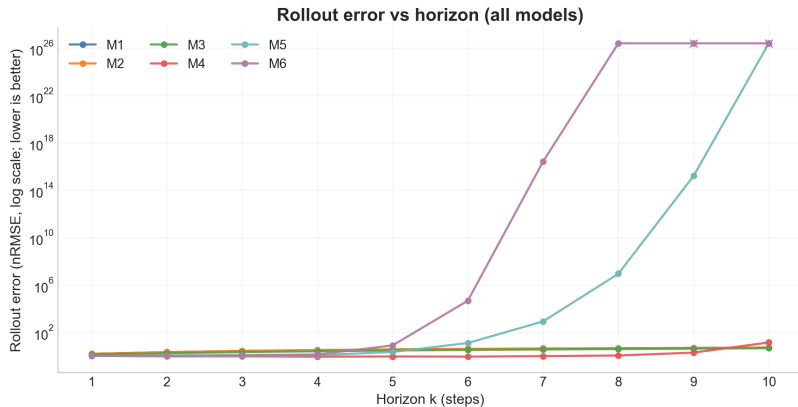
High one-step fit vs stable rollout



Higher R^2 (one-step fit) does not guarantee stable long-horizon rollouts.

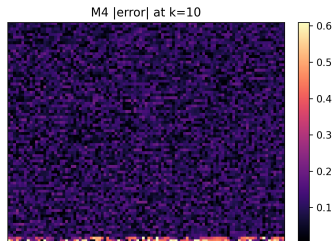
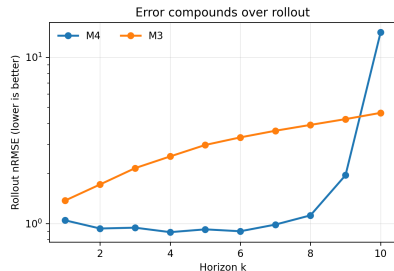
Rollout error vs horizon (all models)

- Curves separate stable vs blow-up models.
- Look at growth rate, not just $k=10$.
- Blow-up = unusable surrogate.



Insight: A stable model keeps error growth controlled as k increases.

Errors compound over rollout



- Error grows roughly multiplicatively.
- Error map shows *where* instability appears.
- Instability often starts localized then spreads.

Insight: Even small one-step errors can accumulate into large rollout divergence.

Summary table (time-test + rollout)

All models are identified via sparse regression (STRidge) on the same candidate library.

Model variant (STRidge)	R^2	One-step RMSE	Rollout nRMSE (k=10)
M3 (+1st-order)	-1.180	136.000×10^{-3}	4.630
M4 ($+u^2$)	459.000×10^{-3}	109.000×10^{-3}	14.100
M4 + translation stab	450.000×10^{-3}	110.000×10^{-3}	10.200
M4 to_first ($\sigma = 2$)	451.000×10^{-3}	109.000×10^{-3}	7.480
M5 (+advection)	275.000×10^{-3}	110.000×10^{-3}	883.000×10^{24}

Best found equation

M4 ($\sigma = 2$): $u_t = 0.3488 u + 0.0073 u_x + 0.0050 u_y - 1.0734 \times 10^{-4} \Delta u - 0.6210 u^2$.

Why: near-best one-step fit (R^2 451.000×10^{-3}) with much better rollout.

- *Interpretation:* M4 wins on R^2 , but rollout is unstable; stabilization improves rollout without improving one-step fit.

Takeaways

Key message

- Separate **short-term fit** from **long-term stability**.
- Identify coefficients via **STRidge**; validate by multi-step rollouts.

What can go wrong

A model can have strong one-step fit yet **diverge** during rollout (error compounding / blow-up).

What works better

With STRidge-based identification, stabilization reduces long-horizon error while leaving one-step fit nearly unchanged; select by rollout first.

Thank You for Your Attention!

Appendix: Adding higher levels of noise to the 2D KS Data

Table: Mean Relative Error: SINDy vs. Weak-SINDy

Noise	SINDy Mean Error	Weak-SINDy Mean Error
1%	5.6%	3.6%
2%	13.4%	8.6%
3%	29.9%	53.6%
4%	79.7%	38.5%
5%	126.4%	37.6%

Low Noise (1–2%):

- Weak-SINDy outperforms standard SINDy, showing the strength of weak formulations.

Transition (3%):

- Both methods struggle; Weak-SINDy shows a temporary error spike.

High Noise (4–5%):

- Standard SINDy fails catastrophically ($> 100\%$ error).
- Weak-SINDy remains relatively stable at $\sim 38\%$.